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Article

Keywords:

Posted Date: September 6th, 2022

DOI: https://doi.org/10.21203/rs.3.rs-2020076/v1

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A sub-wavelength Si LED integrated in a CMOS platform

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(Dated: 22 August 2022)

A nanoscale on-chip light source with high intensity is desired for various applications in integrated photonics systems. However, it is challenging to realize such an emitter using materials and fabrication processes compatible with the standard integrated circuit technology. In this letter, we report an electrically driven Si light-emitting diode (LED) with sub-wavelength emission area fabricated in an open-foundry microelectronics CMOS platform. The LED emission spectrum is centered around 1100 nm and the emission area is smaller than 0.14 μm^2 ($\approx$ ⌀400 nm). This LED has high spatial intensity of > 50 mW/cm^2 which is comparable with the state-of-the-art Si-based emitters with much larger emission areas. Due to sub-wavelength confinement, the emission exhibits a high degree of spatial coherence, which is demonstrated by incorporating the LED into a compact lensless in-line holographic microscope. This centimeter-scale, all-silicon microscope utilizes a single emitter to simultaneously illuminate $\approx$ 9.5 million pixels of a CMOS imager.

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I. INTRODUCTION

In the past two decades, various photonic components, such as couplers, waveguides, and modulators, have been successfully integrated in microelectronic platforms, which enable researchers to translate various bench-top optical systems to photonic chips.[1] Some examples of applications include data communication,[2,3] sensing,[4], imaging[5], and quantum computing.[6] Despite this progress, a small, bright and CMOS compatible on-chip emitter remains elusive and thus in most photonic chips the light originates from off-chip light sources, which leads to low overall energy efficiency and fundamentally limits the scalability of photonic chips.[7,8] Driven by the desire for higher integration density, researchers have fabricated on-chip emitters using various material systems such as rare-earth doped glass,[9,10] Ge-on-Si[11,12] and heterogeneously integrated III-V materials[8]. Emitters based on these materials can have good device performance but the associated fabrication processes are still challenging to integrate in the standard complementary metal-oxide semiconductor (CMOS) platforms.[13] Monolithically integrated native Si is also a promising candidate material for nanoscale and individually controllable emitters due to the nanometer fabrication precision and the large integration scale. However, Si emitters suffer from low quantum efficiency because of the indirect bandgap. This fundamental disadvantage combined with the limitations set by the available materials and fabrication tools hinder the realization of a small, native Si emitter in CMOS.

Tremendous effort has been made to enhance the light emission of Si. Some early work by Green *et al.* focused on using large-area, high quality single crystalline Si with dedicated photon extraction structures.[14,15] These emitters are several cm^2 in area and have intensities of 10^{-4} W/cm^2. These large area emitters rely on the low surface-to-volume ratio to minimize the non-radiative recombination on the surface, and thus it is challenging to reduce the emission area while keeping high efficiency. Other strategies include carrier confinement,[16–23] defect activation,[17,24–26] field emission effect,[27] Purcell enhancement,[23,26,28] and avalanche effect[29–31]. To date, the Si emitter with the smallest emission area as well as the highest intensity was reported by Schmitt *et al.*.[23] In their emitter, the carrier recombination is confined in inverse tapered Si half-ellipsoidal nanostructure which is also an optical cavity. They reported an emission area of approximately $1\mu m^2$ and intensity of 600 mW/cm^2. However, their emitter requires an AFM top contact, which hinders the direct compatibility with CMOS platforms. The smallest emitter fabricated in CMOS was reported in 2021, and employed vertical pn junctions in an unmodified, open-foundry micro-

electronic CMOS node.[32] These devices were scaled to an emission size of 4μm diameter and intensity of over 40 mW/cm^2.

To further scale the emission area down to sub-micrometer dimensions in CMOS platforms, several issues need to be addressed. First, although the bimolecular recombination favors carrier confinement,[33] as the surface-to-volume ratio of the active region increases, the non-radiative recombination assisted by the surface defects (Shockley-Read-Hall recombination, SRH) becomes significant. This is a common problem in most nanoscale opto-electronic devices and might be partially solved by surface passivation.[34] Second, the heat dissipation needs to be efficient to prevent the thermal droop of the quantum efficiency when the current density in the active region is high. As has been reported in Si nano-crystal based emitters, the efficiency droop happens at current densities as low as 1μA/cm^2 and limits the intensity below 1 mW/cm^2.[18,19] Third, the carrier injection structure has to be optimized to support high current while not perturbing light extraction. In our previous work, we have shown that vertical pn junctions with top metal contact can support high injection current while also lower the device footprint compared to lateral junctions.[32] However, the opaque metal contact leads to significant shadowing if the emission area shrinks to a comparable size.

In this work, we report an electrically-driven, nanoscale Si light-emitting diode (LED) realized in an unmodified, open-foundry microelectronic CMOS node. Our LED exhibits sub-wavelength emission area ($< 0.14\mu$m^2, or $\varnothing$400 nm) and high spatial intensity ($>$ 50 mW/cm^2) with the emission spectrum centered around 1100 nm. No efficiency droop is observed at current density up to 2 MA/cm^2. The aforementioned side effects with scaling are mitigated as follows. First, the carriers are spatially confined by the gate oxide layer and a strong electrical field introduced by the nanoscale top contact. This configuration does not rely on extra interfaces except the gate oxide, which is well passivated. Second, carrier recombination happens close to the bulk Si, which conducts heat effectively. Moreover, inspired by recent work on the emission associated with gate oxide breakdown,[35,36] the top contact is fabricated using gate poly-Si instead of metal. The contact is thus transparent and supports further scaling.

Due to sub-wavelength confinement, the emission is spatially coherent. The aperture of the device serves as its own spatial filter. As a potential application, a lensless digital holographic microscope was built using the LED. We demonstrate that the LED is bright enough to simultaneously illuminate 9.5 million pixels ($>$ 1 cm^2 area) of a Si CMOS camera and that the signal-to-noise ratio (SNR) of the holograms is sufficient to reconstruct the images of randomly distributed 20μm

diameter latex beads.

II. DEVICE STRUCTURE AND CHARACTERIZATION

Our LED is fabricated in a 55 nm CMOS node, alongside other functional photonic and electronic components all integrated on the same chip (Fig. 1 (a)). In Fig. 1 (b) (c), we present the schematic of our LED. The LED denoted in the right dashed box in Fig. 1 (b) is a vertical n+/n/p junction. Here the n+, n and p refer to the gate poly-Si layer (degenerate n doped), the n-well (intermediate n doped) and the substrate (lightly p doped), respectively. The thin ($\approx$ 100 nm) gate poly-Si serves as a transparent top contact to the device. The n-well is approximately 1.6 μm in x, 1.8 μm in y and < 2 μm in z. Only a small area (≈ 0.3 μm^2) of the n-well is in contact with the gate oxide ($<$ 3 nm thick). This opening is defined by shallow trench insulator (STI). In this area, a Si filament is formed in the gate oxide by applying electrical stress. This process is usually referred as hard breakdown.[37,38] Microscopically, when high-voltage gate bias is applied, traps form and a conduction path is created. Near this path, the gate current leads to thermal damage which creates more traps. These newly created traps merge into the conduction path which further increases the conductance. This thermal runaway process leads to a point when the Si-O bonds break and the Si in the conduction path melts to form a Si filament. The cross section of the breakdown site is usually sub-100 nm and the Si filament is a resistor in series with the gate poly-Si and the n-well.[38] Similar methods were used to fabricate the contact of the gate poly-Si and the bulk Si substrate by Akil *et al.* and Mihaychuk *et al.*.[35,36] In our test, the substrate was biased at 6.0 V with the poly-Si grounded to introduce gate oxide breakdown. We observed that the current rapidly increased from pA to mA and was stabilized within 1 min. After the breakdown, the IV characteristic was stable and showed no obvious hysteresis.

The carrier transport and recombination process in the LED are illustrated in Fig. 1 (c) in which the gate poly-Si is grounded and the substrate is at positive bias. Due to the negative gate bias, a hole accumulation layer forms near the gate oxide. In the accumulation layer, the holes recombine near the Si filament where electrons are injected from the poly-Si.[35] This process leads to a highly localized emission spot. As the bias increases, the holes are attracted and confined by the increasing electrical field in the Si filament and the emission becomes more localized. In Fig. 1 (d), the micrograph with the LED turned on is presented and the bright spot is observed. As the electrons diffuse into the substrate, they recombine with the majority holes and the associated

emission is spatially broad because the electron diffusion length can be several millimeters in the lightly p doped substrate.[39] The radiative recombination of holes and the majority electrons in the n-well bulk also contribute to the total emission but is insignificant since the size of the n-well ($< 2\mu$m) is much smaller than the diffusion length of the minority holes (10 - 100 μm in intermediate n doped Si).[40] Therefore, the emission in our LED mainly consists of two spatially separate components – a bright localized spot and a faint, spatially broad background.

The emission mechanism is verified by measuring the emission spectra of the LED and a reference emitter, which is fabricated close to the LED in the identical geometry and with similar doping levels but with p-poly-Si gate and a p-well. The results are presented in Fig. 1 (e), where the emission was coupled into a single-mode fiber (SMF) to spatially filter the background. Although the reference emitter also emits photons, the emission spectrum is distinct from the LED. The LED spectrum peak is around 1100 nm, which is associated with the phonon-assisted band-to-band transition. The spectral full-width-half-maximum (FWHM) of the emission increases monotonically from approximately 150 nm to 190 nm (Fig. 1 (f) inset). Compared with the Si LEDs on the micrometer scale, our emission spectrum is $50-150\%$ broader. This is in agreement with the reports on the emission from gate oxide breakdown.[35,36,41–43] The broadening can be attributed to the elevated temperature of the hot carriers accelerated by the electrical field in the Si filament. As the bias increases, the carrier temperature increases and so does the spectral FWHM of the emission. However, the emission peak around 1100 nm shows no red-shift with large bias while the red-shift with increasing temperature is observed in other Si LEDs due to the bandgap shrinking.[14,16] This indicates that the lattice temperature of our LED does not increase significantly because the Si substrate dissipates heat efficiently. As for the reference emitter, although the peak around 1100 nm is observable, the spectrum is much broader than those from the LED. Since the reference emitter (p+/p/p) is uni-polar, the light emission is mainly from impact ionization and hot carrier transition as discussed by Rasras *et al.* and Mihaychuk *et al.*.[36,42] The comparison of the distinct spectra confirms that phonon-assisted band-to-band transition dominates the radiative recombination in the LED while intraband transition and hot-carrier induced transition involving higher energy bands are not significant. The broad spectrum of the reference emitter also serves to validate the broad optical bandwidth of the collection optics and the detectors.

Images of LED emission at various currents are presented in Fig. 2 (a)-(c). The focus position was adjusted such that the signal at the center of the emission was maximized. The emission pattern was characterized by fitting the images into two 2D Gaussian functions. (See supplementary

materials for details.) As an example, the 2D fit result of 6 mA is presented in Fig. 2 (d). The components of the fit in the x and y cross-sections are presented in 2 (f) and (g), respectively. The fit results are in good agreement with the measured data and clearly consist of the two expected components of the emission. At the center of the LED, a bright spot dominates the intensity, while the diffusive background extends ± 10 μm away from the center (Fig. 2 (f) (g) inset). This bright spot is referred as the n+/n emission spot in the following text since the corresponding carrier recombination happens near the n+/n interface.

In Fig. 2 (e), we present the n+/n emission power and the external quantum efficiency (EQE) versus the current. The emission power was evaluated by integrating the Gaussian fit of the spot corrected by the sensitivity of the camera and the transmission of the microscope. (See supplementary S3.) At 6 mA, the n+/n emission power and the EQE are approximately 94.0 pW and 1.4×10^{-8}, respectively. The monotonically increasing EQE with the current indicates the super-linear dependence of the n+/n emission power on the current which is a result of the carrier confinement because the bimolecular recombination rate scales with the product of the excess carrier concentrations. Note that the EQE shows no thermal droop even when the current density is above 2 MA/cm^2 (a conservative estimation using the 0.3 μm^2 opening area as the cross section). This is mainly because the LED is embedded in the Si substrate which is an efficient heat sink.

In Fig. 2 (h) we present the emission area and the spatial intensity of the n+/n spot. The emission area is estimated by deconvolving the spatial FWHM of the emission spot from the Gaussian fit using the independently measured point-spread-function (PSF) of the microscope. (See supplementary S4.) From 1 mA to 6 mA, the emission area shrinks from approximately 0.30 ± 0.12 μm^2 to 0.09 ± 0.04 μm^2. This is consistent with the carrier transport process in which the holes are confined by the electrical field in the Si filament and the localization becomes stronger with the bias increasing. Below 4 mA, the emission area is close to the STI defined opening area (Fig. 1 (b) (c)). This indicates that at low bias the carriers are mainly confined by the STI and thus the emission area does not decrease significantly. The spatial intensity is evaluated using the emission area and the power of the emission spot (Fig. 2 (e)). At 6 mA, the intensity is approximately 102 ± 48 mW/cm^2. Compared with the work on the smallest CMOS emitters,[32] the emission area is 2 orders of magnitude smaller and the average intensity is approximately doubled. The compactness of the emission was further verified by measuring the power coupled into a SMF. In Fig. 2 (i), we present the SMF-coupled power and the SMF coupling efficiency versus the current corrected by the microscope transmission. The SMF-coupled power was maximized by

moving the sample using a piezo stage. During the test, the xyz position corresponding to the maximum power did not change with the current. The SMF coupling efficiency was computed by dividing the SMF-coupled power by the n+/n emission power (Fig. 2 (e)). From 4 mA to 6 mA, the SMF coupling efficiency increases from approximately 40% to 45% while it stays approximately 40% under 4 mA. The trend is consistent with the decrease of the emission area above 4 mA (Fig. 2 (h)) and the high SMF coupling efficiency is due to the sub-wavelength emission area.

Our LED is the smallest Si emitter in CMOS platforms and its intensity is comparable to state-of-the-art Si emitters with much larger areas. This is achieved by our unique design that utilizes field-induced carrier confinement and substrate-assisted heat dissipation. A detailed benchmark of our LED compared with other reported emitters is presented in supplementary S5.

The performance variance of our LEDs on multiple chip dies is not significant. The LEDs are also robust over time if operated below 7.0 mA. After being switched on and off for $\approx 10^5$ times, the SMF-coupled emission power is lowered by approximately 25%. This is probably due to lateral propagation of the breakdown site[37]. To be conservative, all the results presented in this report are from the LEDs which were slightly degraded during the robustness test. (supplementary S6) We also characterized the reference emitter and the results are presented in supplementary S7. In summary, compared with the LED, besides the distinct emission mechanism, the reference emitter exhibits a linear IV curve and much larger resistance (> 20 kΩ). It also has a lower safe current (≈ 0.9 mA). We conclude that the reference emitter is not suitable as a on-chip light source because of the requirement for high bias voltage and its lack of robustness.

III. DIGITAL IN-LINE HOLOGRAPHY

The small emission area of our LED results in highly spatially coherent light which enables interference-based applications including holographic microscopy. We incorporated the LED as a point source for direct illumination in a lensless in-line holography setup (Fig. 3 (a)), by which an interference pattern between the un-scattered incident light and the light scattered by a sample was generated and recorded with a Si CMOS camera of 17.6 mm $\times$ 13.3mm size. We used this setup to image a sample containing 20 μm-diameter latex beads randomly distributed on a glass slide. In Fig. 3 (b) (c) and (d), we present the full hologram on the imager, a close-up of the hologram and the counts on a line segment, respectively. In Fig. 3 (c) (d), the zeroth and the first order bright rings are observed, which verifies that the spatial coherence our LED is high enough

to compensate for the relatively broad spectrum to generate interference patterns. In Fig.3 (b), we denote a white dashed box of ≈ 1.25 cm $\times 1.10$ cm, where the interference pattern of the beads can be clearly observed. This illumination area corresponds to $\mathrm{NA}_x \approx 0.46$ and $\mathrm{NA}_y \approx 0.41$. The high NA of our LED (by virtue of its small emission area) allows us to achieve a large field of view with minimal distance between the source and the sample (1 cm). In conventional in-line holography setups employing LEDs and pinholes such a distance is on the order of 10 cm or above because the illumination NA is limited to $\approx 0.62\lambda/r$ for a circular pinhole with tens μm diameter. The pinhole-free configuration of our setup also simplifies the alignment.

We performed hologram reconstruction using the angular spectrum method[44] - the recorded hologram was transformed to the frequency domain through fast Fourier transform (FFT), back-propagated to the image plane in the frequency domain and then converted back to the space domain with an inverse FFT. The reconstruction result is presented in Fig. 3 (e) (f). Compared with the optical micrograph (Fig. 3 (g)), the clusters of the beads can be clearly resolved from each other and their relative positions are accurate. The reconstruction also exhibits the details of the sample. In Fig. 3 (f) (g), three clusters containing 1 2 and 3 beads are labeled. Their shape and size are distinct. We thus conclude the resolution of our holography is approximately the size of 1 bead ($\approx 20\mu$m). The limitations of this simple reconstruction algorithm include the assumption of a monochromatic source (which is not true in our case) and the presence of the twin image. Such limitations can be overcome by employing more advanced techniques such as iterative phase retrieval[45,46] and machine learning-based reconstruction methods[47]. In our case, machine learning approaches robust to low SNR[48] are particularly attractive.

IV. DISCUSSION AND CONCLUSION

In this report, we present a CMOS-integrated sub-wavelength scale LED exhibiting the smallest emission area (< 0.14 μm^2, $\varnothing$400 nm) among the reported Si emitters and high spatial intensity (> 50 mW/cm^2). In our LED, the heat is dissipated efficiently through the Si substrate so no thermal droop of the EQE is observed even when the current density is above 2 MA/cm^2. The top contact is designed such that a strong electrical field spatially confines the carriers. The only surface that contributes to the confinement at high bias is the gate oxide with high passivation. The top contact is also transparent to avoid shadowing on the emission, which supports further scaling. In the current design, the performance of the LED is limited by the robustness of the top

contact. Given the trend that the EQE and the intensity increases monotonically with the current, the performance of the LED can be further enhanced if the the geometry of the top contact can be optimized to supply a higher current. The substrate p contact can be also fabricated close to the LED to lower the background emission.

As a potential application, we integrated the LED into a in-line holographic microscope. The large NA and the high spatial coherence of our LED combined brought us a centimeter-scale, all-silicon holographic microscope requiring no lens or pinhole. These LEDs can also be arrayed in CMOS to generate programmable coherent illumination for more complex systems in the future. While a simple FFT-based algorithm was used to reconstruct the image, we obtained reasonably good quality results. The resultant resolution of our image is limited by the low SNR of the hologram and the blurring caused by the twin image, a well-known issue of in-line holography. More advanced computational techniques can be used to address both limitations.

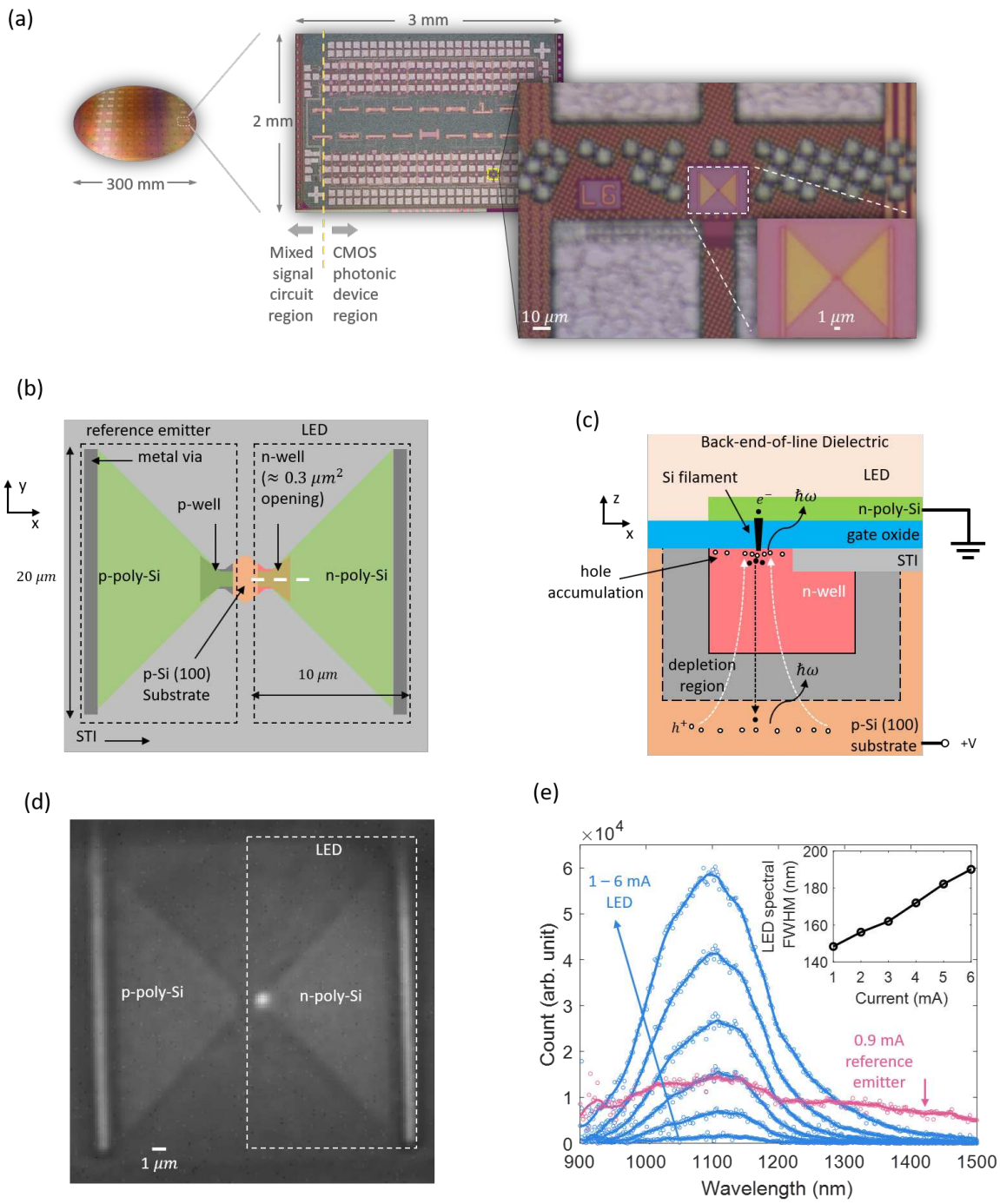
(a)
3 mm
2 mm
300 mm
Mixed signal circuit region
CMOS photonic device region
10 μm
1 μm
(b)
reference emitter
LED
metal via
n-well (≈ 0.3 μm² opening)
p-well
y
x
20 μm
p-poly-Si
n-poly-Si
p-Si (100) Substrate
10 μm
STI
(c)
Back-end-of-line Dielectric
Si filament
LED
z
x
n-poly-Si
gate oxide
STI
hole accumulation
n-well
depletion region
p-Si (100) substrate
+V
(d)
LED
p-poly-Si
n-poly-Si
1 μm
(e)
×10⁴
1 – 6 mA LED
LED spectral FWHM (nm)
Current (mA)
0.9 mA reference emitter
Count (arb. unit)
Wavelength (nm)

FIG. 1: (a) Photograph of a fully fabricated 300 mm wafer with monolithic electronics and photonics, and optical micrograph of a diced, unpackaged chip with different active and passive photonic components and mixed-signal circuits integrated side-by-side, and close-ups of the LED and the reference emitter on this chip. (b) A schematic top view of the LED and the reference emitter. Here the back-end-of-line (BEOL) dielectric and the gate oxide are not shown. (c) A zoom-in side view of the LED on the white dashed line in (b) and the corresponding carrier transport. The solid and hollow circles indicate the electrons and holes, respectively. The white and black dashed arrows indicate the hole and electron transport, respectively. (d) A micrograph of the LED when it is biased at 6 mA. The wide-field illumination light is from a infrared LED and centered around 1100 nm. (e) The spectra of the LED and the reference emitter measured by routing the emission through a single-mode fiber into a spectrometer based on an InGaAs camera. The hollow circles are the raw data and the solid lines are from the Savitzky–Golay filter with polynomial order 3 and frame length 21. The spectral FWHM of the LED versus the current is presented in the inset.

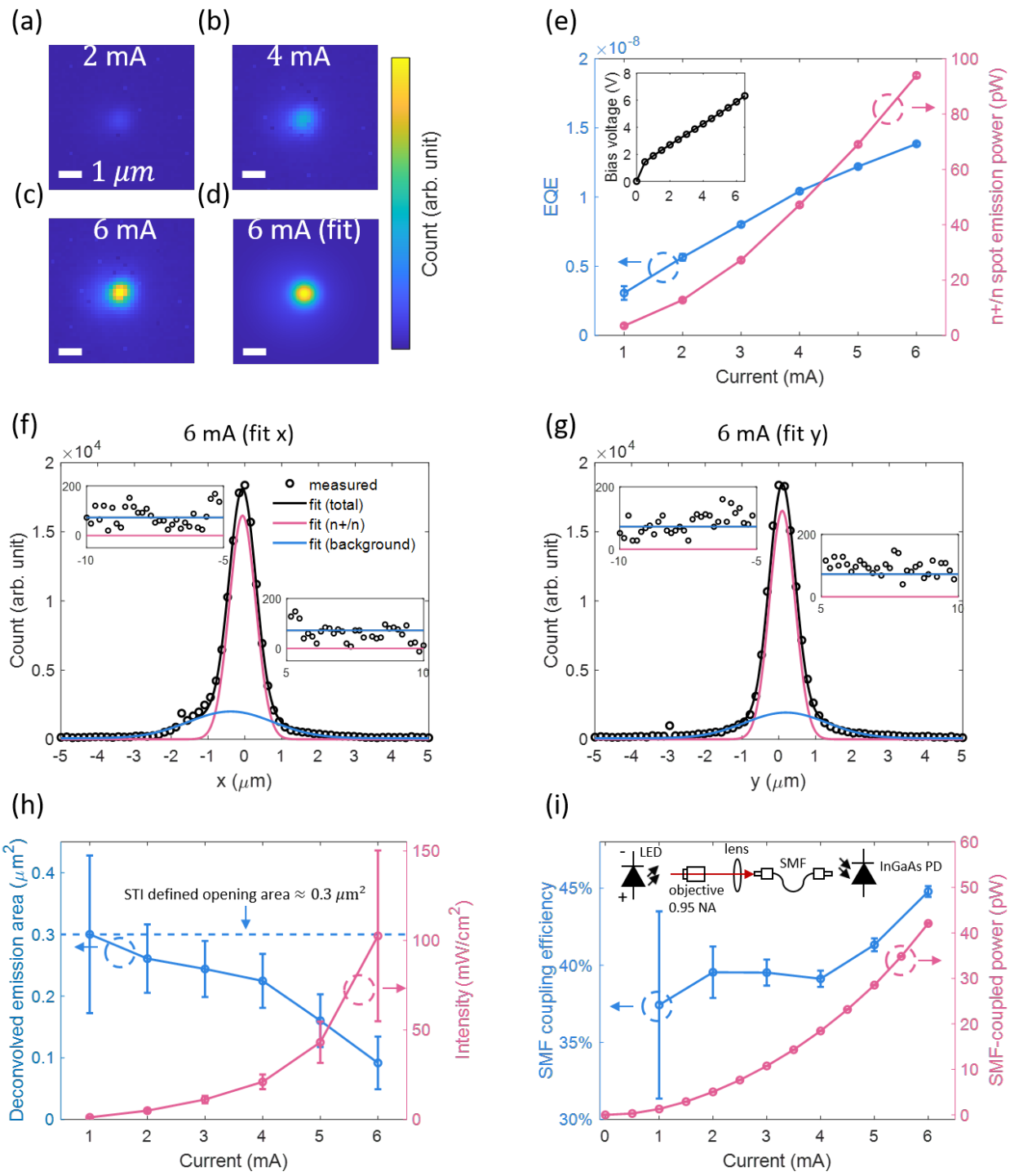
(a)
2 mA
1 μm
(b)
4 mA
(c)
6 mA
(d)
6 mA (fit)
Count (arb. unit)
(e)
×10^-8
EQE
Bias voltage (V)
n+/n spot emission power (pW)
Current (mA)
(f)
6 mA (fit x)
×10^4
measured
fit (total)
fit (n+/n)
fit (background)
Count (arb. unit)
x (μm)
(g)
6 mA (fit y)
×10^4
Count (arb. unit)
y (μm)
(h)
Deconvolved emission area (μm^2)
STI defined opening area ≈ 0.3 μm^2
Intensity (mW/cm^2)
Current (mA)
(i)
LED
lens
SMF
InGaAs PD
objective
0.95 NA
SMF coupling efficiency
SMF-coupled power (pW)
Current (mA)

FIG. 2: (a)-(c) Images of the emission pattern at multiple currents. A microscope equipped with a 100×, 0.95NA objective and an InGaAs camera was used to characterize the devices. (See supplementary S1) The chip die was wirebonded to a chip carrier and the carrier was fixed on a piezo XYZ stage. (d) 2D Gaussian fit of the emission pattern at 6 mA. (e) The n+/n emission power and the associated EQE versus the current. Note that here EQE only accounts for the emission of the spot while the background is neglected. The inset shows the forward bias voltage versus the current. The error bars are from the 95% confident inverval of the corresponding fitting parameters. (f, g) x and y cross-sections of the fit emission pattern at 6 mA from -5 to 5 μm. The inset figures are the fit from ± 5 to ± 10 μm. Only the 20×20 μm^2 area centered at the spot is considered in the fit because the emission outside of this region is shadowed by metal fill required by the process. (h) The deconvolved emission area and the spatial intensity versus the current. The deconvolution results are based on the FWHM of the PSF of the microscope.(784 ± 50 nm in x and 740 ± 46 in y. See supplementary S2 S4) The dashed line indicates the STI defined opening area which confines the carrier when the bias is low. The error bars are from the error propagation considering the PSF measurement error. (i) The single mode fiber (SMF) coupled power and the coupling efficiency versus the current. The schematic setup is presented to indicate the measurement methods. The SMF is PM980 (Thorlabs, Inc.).

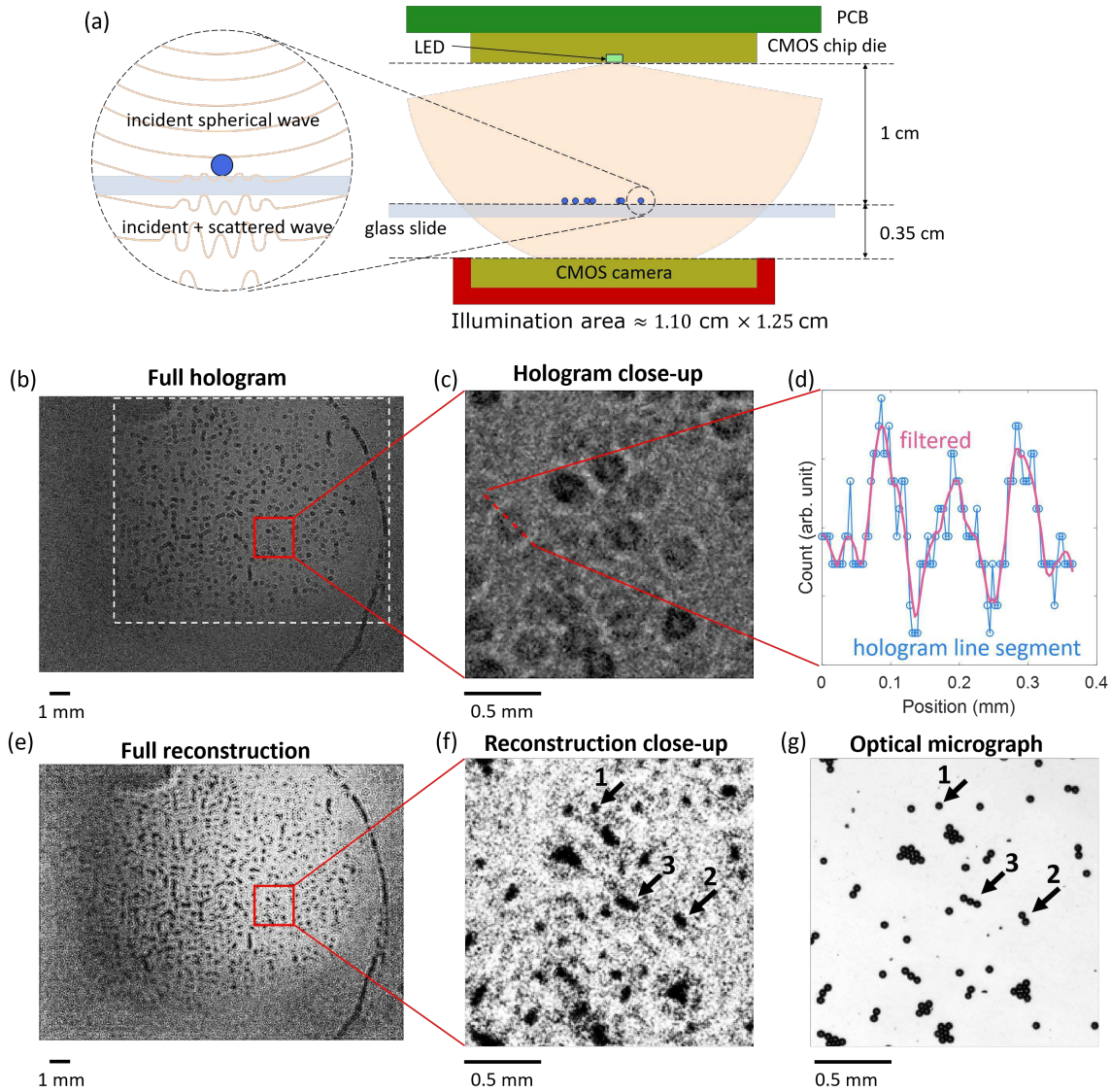
(a)
PCB
LED
CMOS chip die
incident spherical wave
1 cm
incident + scattered wave
glass slide
0.35 cm
CMOS camera
Illumination area ≈ 1.10 cm × 1.25 cm
(b)
Full hologram
1 mm
(c)
Hologram close-up
0.5 mm
(d)
filtered
Count (arb. unit)
hologram line segment
0
0.1
0.2
0.3
0.4
Position (mm)
(e)
Full reconstruction
1 mm
(f)
Reconstruction close-up
1
3
2
0.5 mm
(g)
Optical micrograph
1
3
2
0.5 mm

FIG. 3: (a) Schematic of the experimental setup demonstrating in-line holographic imaging of 20 μm-diameter latex beads with a single LED (bias current at 6.5 mA) as the illumination source. Closeup on the left illustrates the interference pattern formed between the incident light and the scattered light from a bead. (b) The full hologram recorded by a cooled CMOS camera (5 °C) for a 16 s integration. The imager chip size and the pixel pitch are 17.6 mm $\times$ 13.3mm and 3.8 μm, respectively. The raw hologram is filtered by a 5×5 median filter. The white dashed box indicates the illumination area. (c) Close-up of the hologram in the red box in (b). (d) Values of the hologram on the red dashed line in (c). The filtered curve is from Savitzky–Golay filter with polynomial order 3 and frame length 11. (e) Reconstruction of the hologram on the whole imager. The contrast is enhanced by histogram equalization with 64 bins. (f) Close-up of the reconstruction in the same area as (c). (g) Optical micrograph of the sample in the same area as (c) and (f), taken by a 5$\times$ objective.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Z.L. proposed the reported operation mode of the device, built the characterization tools, characterized the device, and drafted the manuscript. J.X. designed the device and performed initial tests in a different operation mode reported elsewhere. M.C. built the digital in-line holography setup and performed the image reconstruction. J.K. assisted J.X. with the tape-out design and the device tests. H.N., D.C., K.Y.L., and E.Q. fabricated the chip. R.R. supervised the project. All authors contributed to the final manuscript.

SUPPLEMENTARY MATERIAL

See supplementary material.

FUNDING

This research is supported by the National Research Foundation (NRF), Prime Minister's Office, Singapore under its Campus for Research Excellence and Technological Enterprise (CREATE) program, Disruptive Sustainable Technology for Agricultural Precision (DiSTAP) and Critical Analytics for Manufacturing Personalized-Medicine (CAMP) are interdisciplinary research groups (IRG) of the Singapore MIT Alliance for Research and Technology (SMART) Centre.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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